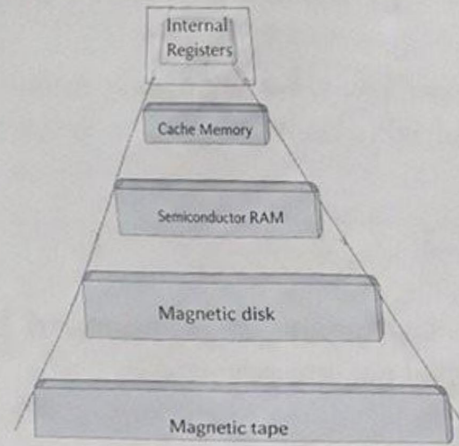


A. Multichoise:



Q1

1. Which level of the memory hierarchy is typically the fastest?

أي مستوى في هرم الذاكرة يكون عادة الأسرع؟

- a) Cache memory
- b) Semiconductor RAM
- ☒ c) Registers
- d) Hard magnetic disks

2. Which level is considered main memory or primary memory?

أي مستوى يُعتبر الذاكرة الرئيسية أو الذاكرة الأساسية؟

- a) Cache memory
- b) Registers
- ☒ c) Semiconductor RAM
- d) Magnetic tape units

20 Marks

3. Which level is considered main memory or primary memory?

أي مستوى يُعتبر الذاكرة الرئيسية أو الذاكرة الأولية؟

- a) Cache memory
- b) Registers
- ☒ c) Semiconductor RAM
- d) Magnetic tape units

4. Which level is part of secondary storage, usually larger in capacity but slower?

أي مستوى يُعد جزءاً من التخزين الثانوي، عادة أكبر سعة ولكن أبطأ؟

- a) Registers
- b) Cache memory
- ☒ c) Hard magnetic disks
- d) Semiconductor RAM

5. Which level is often used for long-term backup or very slow storage?

أي مستوى يستخدم غالباً للنسخ الاحتياطي طويل الأمد أو التخزين البطيء جداً؟

- a) Semiconductor RAM
- b) Cache memory
- ☒ c) Magnetic tape units
- d) Hard magnetic disks

6. As you move down the hierarchy from top to bottom, what generally happens to capacity and access time?

عند النزول في الهرم من الأعلى إلى الأسفل، ماذا يحدث عادة للسعة وزمن الوصول؟

- a) Capacity decreases, access time decreases
- b) Capacity decreases, access time increases
- ☒ c) Capacity increases, access time increases
- d) Capacity increases, access time decreases

7. What is the general goal of having multiple levels in the memory hierarchy?

ما الهدف العام من وجود مستويات متعددة في هرم الذاكرة؟

- a) Use one type of memory to simplify design
- ☒ b) Balance speed, capacity, and cost to improve overall performance
- c) Reduce the number of processors
- d) Make all memory types equally fast at the same cost

8. Which level usually holds more data than cache but is slower than cache?

- a) Registers
- b) Cache memory
- ☒ c) Semiconductor RAM
- d) Magnetic tape units

9. What is the correct order of levels from fastest to slowest?

ما الترتيب الصحيح للمستويات من الأسرع إلى الأبطأ؟

- a) Cache → Registers → Main memory → Secondary memory
- ☒ b) Registers → Cache → Main memory → Secondary memory
- c) Main memory → Registers → Cache → Secondary memory
- d) Secondary memory → Main memory → Cache → Registers

10. Which level typically stores data that the CPU needs immediately or in very short execution cycles?

- a) Magnetic tape units
- b) Hard magnetic disks
- ☒ c) Registers
- d) Semiconductor RAM

Q2 A. Match between the correct answer :

صل ما بين العمود الأول بما يناسبه في العمود الثاني:

(A)	(B)	
Hit Cache	Word, Block	5
Access of method	Sequential, Direct, Random, Associative	2
Miss Cache	Access time, Cycle time, Transfer rate	6
capacity	Number of the words, Number of the bytes	4
Type of Transfer	When the processor finds data in the cache that it is looking for	1
Performance	When the processor looks for data in the cache, but the data is not available it is looking for	3

10 Marks

Answer the following:

1. The design constraints on a computer's memory can be summed up by three questions: How much? How fast? How expensive?

How much -> capacity

How fast -> the access time

How expensive -> cost

Q3

2. what the meaning of (μ op)? and the type of Interrupt?
ما معنى μ op؟ وأنواع Interrupt؟

1. Each of the smaller cycles involves a series of steps, each of which involves the processor registers .

types of Interrupts :

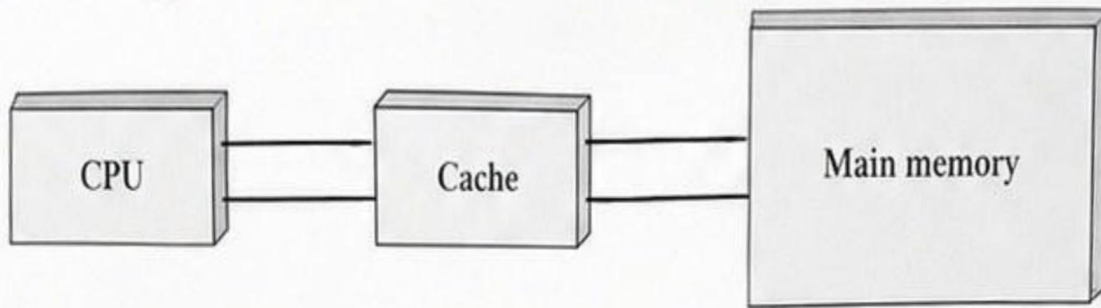
1- Internal Interrupt.

2- Software Interrupt.

3- External Interrupt.

10 Marks

- From this Figure:



A- Fill the blank with the correct answer: املأ الفراغ بالإجابة الصحيحة:

1. The memory located between the CPU and the main memory is called **Cache memory**
الذاكرة الموجودة بين وحدة المعالجة المركزية والذاكرة الرئيسية تسمى
2. In the figure, the transfer between the CPU and the cache is typically done in units of **Word Transfer**
في الشكل، يتم النقل بين المعالج المؤقتة عادةً بوحدات
3. Transfer between the cache and the main memory is usually done in units of **Blocks**
يكون النقل بين الذاكرة والذاكرة الرئيسية عادةً بوحدات

B. Answer the Question: أجب عن الأسئلة:

1. What is the name of the fast memory located between the CPU and the main memory?
ما اسم الذاكرة السريعة الموجودة بين وحدة المعالجة والذاكرة الرئيسية?
Cache memory
2. As you move from faster, smaller memory toward slower, larger memory in a hierarchy, what happens to capacity and access time?
عند الانتقال من ذاكرة أسرع وأصغر إلى ذاكرة أبطأ وأكبر في الهرم، ماذا يحدث للسعة وزمن الوصول?

Capacity increases
Access time increases